Math 314, Exam 2 practice (Spring 2026)

Problem 1. For each of the following vector spaces, indicate its dimension.

(a) $\mathbb{R}^{101}$.

(b) $M_{4\times 5}$, the vector space of all 4×5 matrices.

(c) The null space of $\begin{bmatrix} 3 & 0 & 0 & 9 \\ 0 & 1 & 0 & 8 \\ 0 & 2 & 0 & 16 \end{bmatrix}$.

(d) $\mathbb{P}$, the vector space of all polynomials with real coefficients.

(e) The space of all 4×4 matrices with trace 0.

(f) The space of all 5×5 lower triangular matrices.

Answer 1.

(a) 101

(b) $4 \times 5 = 20$

(c) $\begin{bmatrix} 3 & 0 & 0 & 9 \\ 0 & 1 & 0 & 8 \\ 0 & 2 & 0 & 16 \end{bmatrix} \sim \begin{bmatrix} 3 & 0 & 0 & 9 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ so the dimension of the null space is 2.

(d) ∞.

(e) $12 + 3 = 15$.

Note: there are 4 diagonal entries, but for the trace to be zero, they have to add up to zero. That gives us 3 choices, with the 4th entry completely determined:

$$\begin{bmatrix} a & & & \\ & b & & \\ & & c & \\ & & & -(a+b+c) \end{bmatrix}$$

The remaining entries in the matrix, marked below with a $*$, can be anything:

$$\begin{bmatrix} a & * & * & * \\ * & b & * & * \\ * & * & c & * \\ * & * & * & -(a+b+c) \end{bmatrix}.$$

There are 12 such entries, so the dimension is

$$12 + 3 = 15.$$

(f) $5+4+3+2+1=15$. There are $5+4+3+2+1$ entries below (and including) the main diagonal, which can take any value. Thus the dimension is 15.

Problem 2. Give a basis for each of the following vector spaces.

(a) $\mathbb{R}^3$

(b) $M_{2\times 3}$, the space of 2×3 matrices.

(c) The space of 2×2 upper triangular matrices.

(d) The space of all polynomials of degree up to 3 satisfying $p(0)=0$.

Answer 2.

(a) $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}0\\1\\0\end{bmatrix},\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$.

(b) $\left\{\begin{bmatrix}1&0&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&1&0\\0&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\1&0&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&1&0\end{bmatrix},\begin{bmatrix}0&0&0\\0&0&1\end{bmatrix}\right\}$.

(c) $\left\{\begin{bmatrix}1&0\\0&0\end{bmatrix},\begin{bmatrix}0&1\\0&0\end{bmatrix},\begin{bmatrix}0&0\\0&1\end{bmatrix}\right\}$.

(d) Any such polynomial has the form

$$p(t)=at^3+bt^2+ct$$

(and constant term 0) so a basis is given by

$$\{t,t^2,t^3\}.$$

Problem 3. Show that the set of polynomials

$$W=\left\{bt-bt^2+ct^3\ :\ b,c \text{ any real numbers}\right\}$$

is a subspace of $\mathbb{P}_3$.

Answer 3. Any vector in W is of the form

$$bt-bt^2+ct^3=b(t-t^2)+ct^3$$

with b and c taking any possible real value. Therefore, $W=\operatorname{span} t-t^2,t^3$. Since the span of any set of vectors in a vector space V is a subspace of V, W is a subspace of $\mathbb{P}_3$.

Problem 4. Is the set of vectors

$$\left\{\begin{bmatrix}1&3\\4&6\end{bmatrix},\begin{bmatrix}2&7\\8&14\end{bmatrix},\begin{bmatrix}5&20\\25&40\end{bmatrix},\begin{bmatrix}0&0\\0&1\end{bmatrix}\right\}$$

a basis for $M_{2\times 2}$? Justify your answer.

Answer 4. Consider the following basis for $M_{2\times 2}$:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 01 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

The coordinate vectors for our proposed basis in the basis $\mathcal{B}$ are

$$\begin{bmatrix} 1 \\ 3 \\ 4 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 8 \\ 14 \end{bmatrix}, \begin{bmatrix} 5 \\ 20 \\ 25 \\ 40 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

It's equivalent to decide whether these vectors form a basis for $\mathbb{R}^4$, and since this is a vector space of dimension 4, it's enough to check whether they are a linearly independent set.

$$\begin{bmatrix} 1 & 2 & 5 & 0 \\ 3 & 7 & 20 & 0 \\ 4 & 8 & 25 & 0 \\ 6 & 14 & 40 & 1 \end{bmatrix} \xrightarrow[R3\to R3-4R1]{R2\to R2-3R1} \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 6 & 14 & 40 & 1 \end{bmatrix} \xrightarrow{R4\to R4-6R1} \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 2 & 10 & 1 \end{bmatrix} \xrightarrow{R4\to R4-2R2} \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow[R2\to R2-R3]{R1\to R1-R3} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R3\to \frac{1}{5}R3]{R1\to R1-2R2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There are no free variables, so our vectors are linearly independent, and thus form a basis for $\mathbb{R}^4$. We conclude that the vectors we were given in $M_{2\times 2}$ are also a basis for $M_{2\times 2}$.

Problem 5. Consider the following subset of $\mathbb{R}^4$:

$$W = \left\{ \begin{bmatrix} 4a+3b+3c \\ 0 \\ a+3b \\ 3b-c \end{bmatrix} : \ a, b, c \text{ in } \mathbb{R} \right\}.$$

(a) Show that W is a subspace of $\mathbb{R}^4$.

(b) Find a basis for W.

(c) Find the dimension of W.

Answer 5.

(a) Note that any vector in W is of the form

$$\begin{bmatrix} 4a+3b+3c \\ 0 \\ a+3b \\ 3b-c \end{bmatrix} = a \begin{bmatrix} 4 \\ 0 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 3 \\ 0 \\ 3 \\ 3 \end{bmatrix} + c \begin{bmatrix} 3 \\ 0 \\ 0 \\ -1 \end{bmatrix}.$$

Since a, b, and c can take any value, we see that

$$W = \operatorname{span}\begin{bmatrix}4\\0\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\3\\3\end{bmatrix}, \begin{bmatrix}3\\0\\0\\-1\end{bmatrix}.$$

The span of a set of vectors is a subspace, and thus W is a subspace of $\mathbb{R}^4$.

(b) We showed that

$$W = \operatorname{span}\begin{bmatrix}4\\0\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\3\\3\end{bmatrix}, \begin{bmatrix}3\\0\\0\\-1\end{bmatrix}.$$

Consider

$$A = \begin{bmatrix}4 & 3 & 3\\0 & 0 & 0\\1 & 3 & 0\\0 & 3 & -1\end{bmatrix} \xrightarrow[R_2 \leftrightarrow R_4]{R_1 \leftrightarrow R_3} \begin{bmatrix}1 & 3 & 0\\0 & 3 & -1\\4 & 3 & 3\\0 & 0 & 0\end{bmatrix} \xrightarrow{R_3 \to R_3 - 4R_1} \begin{bmatrix}1 & 3 & 0\\0 & 3 & -1\\0 & -9 & 3\\0 & 0 & 0\end{bmatrix} \xrightarrow{R_3 \to R_3 + 3R_2} \begin{bmatrix}1 & 3 & 0\\0 & 3 & -1\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix}.$$

We see that there is a pivot on the first and second columns, so the first two columns of A form a basis for W. Our basis is

$$\left\{\begin{bmatrix}4\\0\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\3\\3\end{bmatrix}\right\}.$$

(c) 2, since our basis had two elements.

Problem 6. Consider $A = \begin{bmatrix}-3 & 6 & -1 & 1 & -7\\1 & -2 & 2 & 3 & -1\\2 & -4 & 5 & 8 & -4\end{bmatrix} \sim B = \begin{bmatrix}1 & -2 & 2 & 3 & -1\\0 & 0 & 1 & 2 & -2\\0 & 0 & 0 & 0 & 0\end{bmatrix}$.

(a) Find a basis for $\operatorname{col}(A)$.

(b) Find a basis for $\operatorname{nul}(A)$.

(c) Compute the rank of A and the dimension of the null space of A and check that the Rank Theorem holds.

Answer 6.

(a) There are pivots on the first and third column, so

$$\left\{\begin{bmatrix}-3\\1\\2\end{bmatrix}, \begin{bmatrix}-1\\2\\5\end{bmatrix}\right\}$$

is a basis for $\operatorname{col}(A)$.

(b) First, note that

$$B = \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - 2R_2} \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The general solution to $Ax = 0$ is

$$x = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, the following is a basis for $\mathrm{nul}(A)$:

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

(c) Looking at our previous answers, we see that

$$\mathrm{rank}(A) = \dim(\mathrm{col}(A)) = \#\text{pivots of } A = 2$$

and

$$\dim(\mathrm{nul}(A)) = \#\text{free variables} = 3$$

Thus

$$\mathrm{rank}(A) + \dim(\mathrm{nul}(A)) = 2 + 3 = 5$$

and indeed A has 5 columnns.

Problem 7. Let

$$A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix} \text{ and } \mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}.$$

(a) Is $\mathbf{w}$ in $\mathrm{nul}(A)$?

(b) Is $\mathbf{w}$ in $\mathrm{col}(A)$?

Answer 7.

(a) $Aw = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} -16 - 2 + 18 \\ 12 + 4 - 16 \\ -8 - 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$

Yes, w is in $\mathrm{nul}(A)$.

(b) We want to know if the system $Ax = w$ is consistent.

$$\begin{bmatrix} -8 & -2 & -9 & 2 \\ 6 & 4 & 8 & 1 \\ 4 & 0 & 4 & -2 \end{bmatrix} \xrightarrow{R1 \leftrightarrow R3} \begin{bmatrix} 4 & 0 & 4 & -2 \\ 6 & 4 & 8 & 1 \\ -8 & -2 & -9 & 2 \end{bmatrix} \xrightarrow{R1 \to \frac{1}{4}R1} \begin{bmatrix} 1 & 0 & 1 & -\frac{1}{2} \\ 6 & 4 & 8 & 1 \\ -8 & -2 & -9 & 2 \end{bmatrix}$$

$$\xrightarrow[R3 \to R3+8R1]{R2 \to R2-6R1} \begin{bmatrix} 1 & 0 & 1 & -\frac{1}{2} \\ 0 & 4 & 2 & 4 \\ 0 & -2 & -1 & -2 \end{bmatrix} \xrightarrow{R2 \to \frac{1}{4}R2} \begin{bmatrix} 1 & 0 & 1 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} & 1 \\ 0 & -2 & -1 & -2 \end{bmatrix} \xrightarrow{R3 \to R3+2R2} \begin{bmatrix} 1 & 0 & 1 & -\frac{1}{2} \\ 0 & 1 & \frac{1}{2} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are no pivots on the last column, so the system is consistent. Yes, w is in $\mathrm{col}(A)$.

Problem 8. Consider the matrix

$$A = \begin{bmatrix} 0 & 1 & 4 & -5 & 3 \\ 1 & 3 & 5 & -2 & 1 \\ 3 & 7 & 7 & 6 & 1 \\ 4 & 10 & 12 & 4 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -7 & 0 & -34 \\ 0 & 1 & 4 & 0 & -13 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Find bases for the column space and null space of A.

Answer 8. Pivot columns: 1, 2, and 4. Basis for the column space of A:

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 7 \\ 10 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 7 \\ 12 \end{bmatrix} \right\}.$$

The general solution to the system $Ax = 0$ is

$$\begin{cases} x_1 = 7x_3 + 34x_5 \\ x_2 = -4x_3 + 13x_5 \\ x_4 = 2x_5 \\ x_3, x_5 \text{ free variables} \end{cases} \qquad \text{so } x = x_3 \begin{bmatrix} 7 \\ -4 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 34 \\ 13 \\ 0 \\ 2 \\ 1 \end{bmatrix}.$$

The following is a basis for the null space of A:

$$\left\{ \begin{bmatrix} 7 \\ -4 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 34 \\ 13 \\ 0 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

Problem 9. Suppose that A is an 8×8 matrix with a 3-dimensional null space.

(a) What is the dimension of $\mathrm{col}(A)$?

(b) Is A invertible? Why or why not?

Answer 9.

(a) By the Rank–Nullity Theorem,

$$\dim(\mathrm{col}(A)) + \dim(\mathrm{nul}(A)) = 8 \implies \dim(\mathrm{col}(A)) = 8 - 3 = 5.$$

(b) No. The matrix A is invertible if and only if $\mathrm{nul}(A)$ is trivial, but in this case $\dim(\mathrm{nul}(A)) = 3 > 0$.

Problem 10. Suppose that B is a 4×6 matrix. What is the smallest possible dimension for $\mathrm{nul}(B)$? Explain your answer.

Answer 10. The Rank–Nullity Theorem says that

$$\dim(\mathrm{col}(B)) + \dim(\mathrm{nul}(B)) = 6 \implies \dim(\mathrm{col}(B)) = 6 - \dim(\mathrm{nul}(A)).$$

The column space of B is a subspace of $\mathbb{R}^4$, so the dimension of the null space of B satisfies $0 \leqslant \dim(\mathrm{col}(B)) \leqslant 4$. By the Rank–Nullity Theorem,

$$\dim(\mathrm{nul}(A)) = 6 - \dim(\mathrm{col}(A)).$$

Thus

$$2 = 6 - 4 \leqslant \dim(\mathrm{nul}(A)) \leqslant 6 - 0 = 6.$$

We conclude that the smallest possible $\dim(\mathrm{nul}(A))$ is 2.

Problem 11. Show that the first three powers of $1 + t$, namely the polynomials

$$p_1 = 1, \quad p_2 = 1 + t, \quad p_3 = (1 + t)^2 = 1 + 2t + t^2$$

form a basis for the vector space $\mathbb{P}_2$ of polynomials of degree at most 2.

Answer 11. Consider the standard basis

$$\mathcal{B} = \{1, t, t^2\}$$

of $\mathbb{P}_2$. The coordinate vectors of p_1, p_2, and p_3 in the basis $\mathcal{B}$ are

$$[p_1]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad [p_2]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad [p_3]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Thus it is equivalent to show that

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad v_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

form a basis for $\mathbb{R}^3$. To show that, note that

$$\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

has a pivot in every column and every row (in positions (1,1), (2,2), and (3,3)), so v_1, v_2, v_3 form a basis for $\mathbb{R}^3$. We conclude that $\{p_1, p_2, p_3\}$ form a basis for $\mathbb{P}_2$.

Problem 12. Is the set of vectors

$$\left\{ \begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}, \begin{bmatrix} 2 & 7 \\ 8 & 14 \end{bmatrix}, \begin{bmatrix} 5 & 20 \\ 25 & 40 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

a basis for $M_{2\times 2}$? Justify your answer.

Answer 12. Consider the following basis for $M_{2\times 2}$:

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

The coordinate vectors for our proposed basis in the basis $\mathcal{B}$ are

$$\begin{bmatrix} 1 \\ 3 \\ 4 \\ 6 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 8 \\ 14 \end{bmatrix}, \begin{bmatrix} 5 \\ 20 \\ 25 \\ 40 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

It's equivalent to decide whether these vectors form a basis for $\mathbb{R}^4$, and since this is a vector space of dimension 4, it's enough to check whether they are a linearly independent set.

$$\begin{bmatrix} 1 & 2 & 5 & 0 \\ 3 & 7 & 20 & 0 \\ 4 & 8 & 25 & 0 \\ 6 & 14 & 40 & 1 \end{bmatrix} \xrightarrow[R3\to R3-4R1]{R2\to R2-3R1} \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 6 & 14 & 40 & 1 \end{bmatrix} \xrightarrow{R4\to R4-6R1} \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 2 & 10 & 1 \end{bmatrix} \xrightarrow{R4\to R4-2R2} \begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow[R2\to R2-R3]{R1\to R1-R3} \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow[R3\to \frac{1}{5}R3]{R1\to R1-2R2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There are no free variables, so our vectors are linearly independent, and thus form a basis for $\mathbb{R}^4$. We conclude that the vectors we were given in $M_{2\times 2}$ are also a basis for $M_{2\times 2}$.

Problem 13. Let V be a vector space of dimension 3, and let

$$\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3) \text{ and } \mathcal{C} = (\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3)$$

be two bases for V such that

$$\mathbf{b}_1 = \mathbf{c}_1 - 3\mathbf{c}_2 + 2\mathbf{c}_3, \mathbf{b}_2 = \mathbf{c}_2 - 3\mathbf{c}_3, \text{ and } \mathbf{b}_3 = -2\mathbf{c}_1 + 4\mathbf{c}_2 + 4\mathbf{c}_3.$$

Let $\mathbf{v} = 2\mathbf{b}_1 - \mathbf{b}_3$. Find $[\mathbf{v}]_{\mathcal{C}}$.

Answer 13. Since we already have the basis elements in $\mathcal{B}$ written in terms of the basis elements in $\mathcal{C}$, we know that

$$[\mathbf{b}_1]_{\mathcal{C}} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, [\mathbf{b}_2]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix}, [\mathbf{b}_3]_{\mathcal{C}} = \begin{bmatrix} -2 \\ 4 \\ 4 \end{bmatrix}.$$

so the change of basis matrix from $\mathcal{B}$ to $\mathcal{C}$ is

$$\begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}.$$

Therefore,

$$[\mathbf{v}]_{\mathcal{C}} = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix} [v]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 - 1 \cdot (-2) \\ 0 \\ 2 \cdot 2 - 1 \cdot 4 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}.$$

Problem 14. Find the determinant of

$$A = \begin{bmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -4 & -3 & 5 \\ 2 & 0 & 3 & 5 \end{bmatrix}.$$

Answer 14. The second row is an excellent choice for cofactor expansion, since 3 of the entries are 0. By expanding along the second row, we get

$$\det(A) = -0 + 0 - 3 \cdot \begin{vmatrix} 1 & -2 & 2 \\ 2 & -4 & 5 \\ 2 & 0 & 5 \end{vmatrix} + 0.$$

To compute the determinant of this 3×3 matrix, let's expand along the third row:

$$\det(A) = -3\,(2(-2 \cdot 5 - 2(-4)) - 0 + 5(-4 + 4)) = -3 \cdot 2(-10 + 8) = 12.$$

Problem 15. Suppose that

$$\det \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ r & s & t & u \\ x & y & z & w \end{bmatrix} = 4 \quad \text{and} \quad B = \begin{bmatrix} r & s & t & u \\ e + 2a & f + 2b & g + 2c & h + 2d \\ a & b & c & d \\ x - 3r & y - 3s & z - 3t & w - 3u \end{bmatrix}.$$

What is $\det(-2B)$?

Answer 15. Since B is a 4×4 matrix, $\det(-2B) = (-2)^4 \det(B) = 16 \det(B)$. On the other hand, B is obtained from a matrix of determinant 4 by switching two rows (which multiplies the determinant by -1) and by adding a multiple of a row to another row (multiple times), which doesn't change the determinant. So $\det(B) = -4$. Therefore, $\det(-2B) = 16 \det(B) = 16 \cdot (-4) = -64$.

Problem 16. Let A be an $n \times n$ matrix. If $A \sim C$ and $\det(C) = 3$, must A be invertible? Justify your answer.

Answer 16. Since $\det(C) = 3 \neq 0$, C is invertible. So $C \sim I$. On the other hand, since $A \sim C$ we also have $A \sim I$, so A is also invertible.

Problem 17. Is $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$? If so, what is the corresponding eigenvalue?

Answer 17. Note that

$$\begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2+4+3 \\ 4-6-3 \\ -3+3+1 \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 1 \end{bmatrix}.$$

This is not a multiple of $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$, so $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ is not an eigenvector of this matrix.

Problem 18. Find the eigenvalues of

$$\begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}.$$

Answer 18. We need to find the solutions to the characteristic equation of this matrix, which is

$$\begin{aligned} \begin{vmatrix} 3-\lambda & -2 \\ 2 & -1-\lambda \end{vmatrix} &= 0 \\ (3-\lambda)(-1-\lambda) + 4 &= 0 \\ -3 - 3\lambda + \lambda + \lambda^2 + 4 &= 0 \\ \lambda^2 - 2\lambda + 1 &= 0 \end{aligned}$$

so

$$\lambda = \frac{-(-2) \pm \sqrt{4 - 4 \cdot 1}}{2} = \frac{2 \pm 0}{2} = 1.$$

There is only one eigenvalue: 1.

Problem 19. Let

$$A = \begin{bmatrix} 4 & 0 & -1 \\ 0 & 4 & -1 \\ 1 & 0 & 2 \end{bmatrix}.$$

(a) Find all the eigenvalues of A and their algebraic multiplicities.

(b) Find the eigenspace corresponding to the eigenvalue $\lambda = 3$ for A.

(c) Without performing any additional computations decide whether A is invertible. Explain your answer.

Answer 19.

(a) The characteristic equation of A is

$$\begin{aligned}
\det(A - \lambda I) &= \begin{vmatrix} 4-\lambda & 0 & -1 \\ 0 & 4-\lambda & -1 \\ 1 & 0 & 2-\lambda \end{vmatrix} \\
&= (4-\lambda)^2(2-\lambda) + 0 + 0 - (1 \cdot (4-\lambda)(-1) + 0 + 0) \\
&= (4-\lambda)^2(2-\lambda) + (4-\lambda) \\
&= (4-\lambda)((4-\lambda)(2-\lambda) + 1) \\
&= (4-\lambda)(8 + \lambda^2 - 6\lambda + 1) \\
&= (4-\lambda)(\lambda^2 - 6\lambda + 9).
\end{aligned}$$

Note that

$$\lambda^2 - 6\lambda + 9 = 0 \iff \lambda = \frac{6 \pm \sqrt{(-6)^2 - 4 \cdot 9}}{2} \iff \lambda = \frac{6 \pm \sqrt{0}}{2} = 3.$$

Thus the characteristic polynomial of A is

$$(4-\lambda)(\lambda-3)^2$$

and the eigenvalues of A are 4 with multiplicity 1 and 3 with multiplicity 2.

(b) We need to find the null space of $A - 3I$:

$$A - 3I = \begin{bmatrix} 4-3 & 0 & -1 \\ 0 & 4-3 & -1 \\ 1 & 0 & 2-3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The general solution to the homogeneous system $(A - 3I)x = 0$ is

$$x = \begin{bmatrix} x_3 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and thus the following is a basis for the eigenspace of A associated to 3:

$$\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}\right\}$$

(c) We see that 0 is not an eigenvalue of A, and thus $\det(A) \neq 0$ and A is invertible.

Problem 20. Suppose that λ is an eigenvalue for a matrix C. Show that λ^2 is an eigenvalue for C^2.

(*Hint:* Use the fact that $C\mathbf{v} = \lambda\mathbf{v}$ to compute $C^2\mathbf{v}$. Careful: squaring both sides does not make sense!)

Answer 20. Suppose that λ is an eigenvalue for C. Then there is a nonzero vector v such that

$$Cv = \lambda v.$$

Then

$$C^2v = C(Cv) = C(\lambda v) = \lambda(Cv) = \lambda(\lambda v) = \lambda^2 v.$$

Therefore, the equation $C^2x = \lambda^2 x$ has a nontrivial solution $x = v$, and λ^2 is an eigenvalue of C^2.

Problem 21. Fill in the following statements in a way that makes them be true and provide brief justifications.

(a) If V is a vector space, $\dim(V) = n$ and $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p)$ is linearly independent, then p____n.

(b) If V is a vector space, $\dim(V) = n$, and $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p) = V$, then p____n.

Answer 21.

(a) We must have $p \leqslant n$.

A set of more than n vectors in a space of dimension n is always linearly dependent.

(b) We must have $p \geqslant n$.

A set of less than n vectors cannot span a space of dimension n.

Problem 22. True or false? Provide brief justification if true or an example to show that it is false.

(a) If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in a vector space V such that none of these vectors is a multiple of another, then the list $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ must be linearly independent.

(b) A list of vectors in a vector space V containing repetitions (such as $(\mathbf{v}_1, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p)$) must be linearly dependent.

Answer 22.

(a) False. For example, consider the vectors

$$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \text{and } w = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

None of them is a multiple of another, and yet $w = u + v$, so the list (u, v, w) is linearly dependent.

(b) True. In that case, one of the vectors (the repeated one) is a multiple of another (the same vector), so the list is linearly dependent.

Problem 23. Is it possible for the solution set of a homogeneous linear system of 2000 equations in 2026 variables to be contained in a line? Justify your answer.

Answer 23. No. The coefficient matrix A of such a system has 2000 rows and 2026 columns. The dimension of the solution set of the homogeneous system is the dimension of the null space of A. By the Rank-Nulity theorem,

$$\dim(\mathrm{nul}(A)) + \dim(\mathrm{col}(A)) = 2026 \implies \dim(\mathrm{nul}(A)) = 2026 - \dim(\mathrm{col}(A)).$$

On the other hand, the column space of A is a subspace of $\mathbb{R}^{2000}$, so it has dimension at most 2000. Therefore,

$$\dim(\mathrm{nul}(A)) = 2026 - \dim(\mathrm{col}(A)) \geqslant 2026 - 2000 = 26.$$

In particular, it is impossible for the solution set to be contained in a line, which would mean it has dimension at most 1.

Problem 24. Let A and B be 5×5 matrices such that $\det(AB) = 17$. Must A be invertible? Justify your answer.

Answer 24. We know that A is invertible if and only if $\det(A) \neq 0$. If A were not invertible, then $\det(A) = 0$ and

$$\det(AB) = \det(A)\det(B) = 0 \cdot \det(B) = 0.$$

But $\det(AB) = 17 \neq 0$, so $\det(A) \neq 0$ and A must be invertible.

Problem 25. Is the list of vectors

$$(1 + t^2, t + t^2, 1 + 2t)$$

a basis for $\mathbb{P}_2$? Justify your answer. If this set is not a basis for $\mathbb{P}_2$, does it satisfy any of the conditions for a basis?

Answer 25. Since $\mathbb{P}_2$ is a vector space of dimension 3, and we have 3 vectors, it's sufficient to check if these vectors are linearly independent. Using the standard basis $\mathcal{B} = \left\{1, t, t^2\right\}$ for $\mathbb{P}_2$, it's sufficient to check if the coordinate vectors

$$[1 + t^2]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, [t + t^2]_{\mathcal{B}} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \text{ and } [1 + 2t]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

are linearly independent.

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 0 \end{bmatrix} \xrightarrow{R3 \to R3 - R1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{bmatrix} \xrightarrow{R3 \to R3 - R2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{bmatrix}$$

There are no free variables, so these three vectors in $\mathbb{R}^3$ are linearly independent, and thus so are our original vectors in $\mathbb{P}_2$. We conclude that the given set of vectors is indeed a basis for $\mathbb{P}_2$.

Problem 26. Let $T\colon \mathbb{P}_1 \to \mathbb{P}_2$ be a linear transformation with the following properties:

$$T(2) = 4 - 2t + 6t^2 \quad \text{and} \quad T(1+t) = 3 + 2t + 6t^2.$$

Find the matrix representing T in the basis

$$\mathcal{B} = (1, t) \text{ for } \mathbb{P}_1 \quad \text{and} \quad \mathcal{C} = (1, t, t^2) \text{ for } \mathbb{P}_2.$$

Answer 26. We need to find $T(1)$ and $T(t)$. We have

$$T(1) = T\left(\tfrac{1}{2} \cdot 2\right) = \tfrac{1}{2} T(2) = \tfrac{1}{2}\left(4 - 2t + 6t^2\right) = 2 - t + 3t^2$$

and

$$T(t) = T((1+t) - 1) = T(1+t) - T(1) = 3 + 2t + 6t^2 - (2 - t + 3t^2) = 3t^2 + 3t + 1.$$

Finally, we need to write the coordinate vectors for $T(1)$ and $T(t)$ in the basis $\mathcal{C}$:

$$[T(1)]_{\mathcal{C}} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad \text{and} \quad [T(t)]_{\mathcal{C}} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}.$$

Therefore, the matrix representing T in the bases $\mathcal{B}$ and $\mathcal{C}$ is

$$\begin{bmatrix} 2 & 1 \\ -1 & 3 \\ 3 & 3 \end{bmatrix}$$

Problem 27. Suppose that $\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ and $\mathcal{C} = (\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3)$ are two bases for $\mathbb{R}^3$.

(a) Explain why $\mathcal{B}' = (\mathbf{b}_2, 2\mathbf{b}_3, \mathbf{b}_1)$ and $\mathcal{C}' = (\mathbf{c}_3, \mathbf{c}_2, \mathbf{c}_1)$ are also bases for $\mathbb{R}^3$. (*Hint:* Row operations.)

(b) Suppose that $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ has matrix representation relative to $\mathcal{B}$ and $\mathcal{C}$ given by

$$[T]_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} 0 & -1 & 2 \\ 3 & 1 & 4 \\ \pi & 0 & -7 \end{bmatrix}.$$

Find $[T]_{\mathcal{C}' \leftarrow \mathcal{B}'}$, the matrix representation of T relative to $\mathcal{B}'$ and $\mathcal{C}'$.

Answer 27.

(a) Since $\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$ is a basis, the matrix with these vectors as rows is invertible. The list $\mathcal{B}' = (\mathbf{b}_2, 2\mathbf{b}_3, \mathbf{b}_1)$ is obtained by permuting and scaling one vector by a nonzero scalar. These correspond to elementary row operations, which preserve invertibility. Hence $\mathcal{B}'$ is also a basis. The same argument applies to $\mathcal{C}' = (\mathbf{c}_3, \mathbf{c}_2, \mathbf{c}_1)$.

(b) We are given the following information about coordinate vectors.

$$[T(\mathbf{b}_1)]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 3 \\ \pi \end{bmatrix}, \quad [T(\mathbf{b}_2)]_{\mathcal{C}} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad [T(\mathbf{b}_3)]_{\mathcal{C}} = \begin{bmatrix} 2 \\ 4 \\ -7 \end{bmatrix}.$$

This means:

$$\begin{aligned} T(\mathbf{b}_1) &= 0\mathbf{c}_1 + 3\mathbf{c}_2 + \pi\mathbf{c}_3, \\ T(\mathbf{b}_2) &= -1\mathbf{c}_1 + 1\mathbf{c}_2 + 0\mathbf{c}_3, \\ T(\mathbf{b}_3) &= 2\mathbf{c}_1 + 4\mathbf{c}_2 - 7\mathbf{c}_3. \end{aligned}$$

Rearrange and scale to get the coordinate vectors of $T(\mathbf{b}_2)$, $T(2\mathbf{b}_3)$, and $T(\mathbf{b}_1)$ in the basis $\mathcal{C}'$:

$$\begin{aligned} T(\mathbf{b}_2) &= -1\mathbf{c}_1 + 1\mathbf{c}_2 + 0\mathbf{c}_3 = 0\mathbf{c}_3 + 1\mathbf{c}_2 - 1\mathbf{c}_1, \\ T(2\mathbf{b}_3) = 2T(\mathbf{b}_3) &= 4\mathbf{c}_1 + 8\mathbf{c}_2 - 14\mathbf{c}_3 = -14\mathbf{c}_3 + 8\mathbf{c}_2 + 4\mathbf{c}_1, \\ T(\mathbf{b}_1) &= 0\mathbf{c}_1 + 3\mathbf{c}_2 + \pi\mathbf{c}_3 = \pi\mathbf{c}_3 + 3\mathbf{c}_2 + 0\mathbf{c}_1. \end{aligned}$$

So we have the coordinate vectors we need:

$$[T(\mathbf{b}_2)]_{\mathcal{C}'} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \quad [T(2\mathbf{b}_3)]_{\mathcal{C}'} = \begin{bmatrix} -14 \\ 8 \\ 4 \end{bmatrix}, \quad [T(\mathbf{b}_1)]_{\mathcal{C}'} = \begin{bmatrix} \pi \\ 3 \\ 0 \end{bmatrix}.$$

These are the columns of the matrix we want:

$$[T]_{\mathcal{C}' \leftarrow \mathcal{B}'} = \begin{bmatrix} | & | & | \\ [T(\mathbf{b}_2)]_{\mathcal{C}'} & [T(2\mathbf{b}_3)]_{\mathcal{C}'} & [T(\mathbf{b}_1)]_{\mathcal{C}'} \\ | & | & | \end{bmatrix} = \begin{bmatrix} 0 & -14 & \pi \\ 1 & 8 & 3 \\ -1 & 4 & 0 \end{bmatrix}.$$

Problem 28. Give an example or explain why no example exists.

(a) A 2×2 matrix A with eigenvalues -1, 2, and 3.

(b) A 3×3 matrix A with eigenvalues -1, 2, and 3.

(c) A 4×4 matrix A with no (real) eigenvalues.

(d) A 3×3 matrix A with no (real) eigenvalues.

(e) A 314×314 matrix A with $\det(A) = 0$ and no (real) eigenvalues.

(f) An upper-triangular 4×4 matrix A with no (real) eigenvalues.

Answer 28.

(a) No example exists. A 2×2 matrix can have at most 2 eigenvalues.

(b) There are many correct examples; a diagonal matrix is easiest:

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

(c) Remember a 2×2 example from class: the matrix $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ has characteristic polynomial $\lambda^2 + 1$, which has no real roots. We can use this matrix as a block to build a 4×4 matrix with no real eigenvalues:

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}.$$

You can check that this 4×4 matrix has characteristic polynomial $(\lambda^2 + 1)^2$, which has no real roots.

(d) Impossible. A 3×3 matrix has a degree-3 characteristic polynomial. Every polynomial of odd degree with real coefficients has at least one real root, by calculus! The limit is $+\infty$ in one direction and $-\infty$ in the other, so by the Intermediate Value Theorem there must be a root somewhere.

(e) Impossible. If $\det(A) = 0$, then A is non-invertible, so 0 is an eigenvalue of A.

(f) Impossible. The eigenvalues of an upper-triangular matrix are the entries on the main diagonal, so an upper-triangular matrix must have at least one real eigenvalue.

Problem 29. Indicate whether each statement is true or false, and justify your answer.

(a) A square matrix A is invertible if and only if 0 is not an eigenvalue of A.

(b) A square matrix A has determinant 0 if and only if 0 is an eigenvalue of A.

(c) If A and B are 314×314 matrices each with eigenvalue 7, then $A + B$ has eigenvalue 7.

(d) If $[T]_{\mathcal{B} \leftarrow \mathcal{B}} = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is the matrix of a linear transformation $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ relative to a basis $\mathcal{B} = (\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3)$, then $T(\mathbf{b}_1 + \mathbf{b}_2 + \mathbf{b}_3) = 8\mathbf{b}_1 + 2\mathbf{b}_2 + \mathbf{b}_3$.

(e) There exists a matrix with exactly 100 eigenvectors.

(f) The matrices $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, and $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ form a basis for the space of symmetric 2×2 matrices.

(g) There is a surjective linear transformation $T\colon \mathbb{R}^{314} \to \mathbb{R}^4$ with $\dim(\ker(T)) = 100$.

(h) If A and B are row-equivalent, and π is an eigenvalue of A, then π must be an eigenvalue of B.

(i) If A and B are row-equivalent, and 0 is an eigenvalue of A, then 0 must be an eigenvalue of B.

Answer 29.

(a) True. A square matrix A is invertible if and only if $\mathrm{nul}(A) = \mathrm{nul}(A - 0I)$ contains only the zero vector, iff 0 is not an eigenvalue of A.

(b) True. A square matrix A has determinant 0 if and only if A is non-invertible, iff 0 is an eigenvalue of A.

(c) False. For example, let $A = B = 7I_{314}$. Then $A + B = 14I_{314}$ has characteristic polynomial $(\lambda - 14)^{314}$, so 7 is not an eigenvalue of $A + B$.

(d) True! We have

$$\begin{aligned} T(\mathbf{b}_1 + \mathbf{b}_2 + \mathbf{b}_3) &= T(\mathbf{b}_1) + T(\mathbf{b}_2) + T(\mathbf{b}_3) \\ &= (3\mathbf{b}_1 + 0\mathbf{b}_2 + 0\mathbf{b}_3) + (1\mathbf{b}_1 + 2\mathbf{b}_2 + 0\mathbf{b}_3) + (4\mathbf{b}_1 + 0\mathbf{b}_2 + 1\mathbf{b}_3) \\ &= 8\mathbf{b}_1 + 2\mathbf{b}_2 + \mathbf{b}_3. \end{aligned}$$

(e) False! If a matrix has one eigenvector relative to λ, then it must have infinitely many. Remember that the λ-eigenspace of A is a subspace.

(f) True. They span:

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

And they are linearly independent:

$$\begin{aligned} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} &= x \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + y \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} + z \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ \implies \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} &= \begin{bmatrix} x & z \\ z & y \end{bmatrix} \\ \implies x = 0, y = 0, z &= 0. \end{aligned}$$

(g) False. If $T\colon \mathbb{R}^{314} \to \mathbb{R}^4$ is surjective, then the standard matrix A of T has rank 4. By the Rank–Nullity Theorem, $\dim(\mathrm{nul}(A)) = 314 - 4 = 310 \neq 100$. (In fact, regardless of whether T is surjective, we must have $\dim(\mathrm{nul}(A)) \geq 314 - 4 = 310$.)

(h) False. Consider

$$A = \begin{bmatrix} \pi & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then A and B are row-equivalent (by one scaling), and π is an eigenvalue of A, but π is not an eigenvalue of B.

(i) True. If A and B are row-equivalent, then the linear systems $Ax = 0$ and $Bx = 0$ have exactly the same solution set (remember that this was the whole point of defining row-equivalence). So $Ax = 0$ has a nontrivial solution (iff 0 is an eigenvalue of A) iff $Bx = 0$ has a nontrivial solution (iff 0 is an eigenvalue of B).